CAMBRIDGE
International Education

Cambridge O Level

PHYSICS **5054 / 42**

Paper 4 Alternative to Practical **May/June 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

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[Turn over

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct / valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	information missing or insufficient for credit
	benefit of the doubt given
	contradiction in response, mark not awarded
	incorrect point or mark not awarded
	error carried forward applied
	used to highlight parts of an extended response
	highlight
	incorrect or insufficient point ignored while marking the rest of the response
	two statements are linked
	power of ten error
	special case **no**

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Annotation	Meaning
SEEN	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
T	answer outside the tolerance of the mark scheme
	correct point or mark awarded
TV	response is too vague or there is insufficient detail in response
U	incorrect unit
AE	evaluation attempted.

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Mark Scheme Abbreviations:

;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
underline	actual word given must be used by candidate (grammatical variants excepted)
( )	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
mp	marking point
cao	Correct answer only

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Question	Answer	Marks	Guidance
1	**PUT 'SEEN' ON THE 3 PAGES ABOVE 1(a)(i) AND LINK TO RELEVANT QUESTIONS, IF APPROPRIATE**		
1(a)(i)	l = 7.0 (cm)	**1**	correct answer only, not 7 or 7.00
1(a)(ii)	0.35 (0)(m) / candidate's **(a)(i)** × 0.05	**1**	
1(b)(i)	23.8 (s)	**1**	correct answer only
1(b)(ii)	1.19 / candidate's **(b)(i)** ÷ 20	**1**	3 significant figures only, correctly rounded
1(c)	doubling / does **not** double T OR T / l is not constant OR l / T is not constant	**1**	no need for any working to be shown
1(d)(i)	g_1 correct (9.6) and to 2 significant figures	**1**	**ecf** (a)(ii) and / or (b)(ii)
1(d)(ii)	9.4 / 9.45 / 9.446 (m/s^2)	**1**	correct answer only, any significant figures >1
1(e)	statement matches results (expect YES) correct calculation to support the statement using both values of g	**1**	**ecf (d)** i.e., values used to show that the difference between the values is < 10 % of either, or the ratio of the values is ⩾ 0.9 (90 %) a correct calculation, but an incorrect statement can get the second mark if both values are identical then the statement 'the values are the same' scores both marks, without the need for a calculation
1(f)	rule placed close to pendulum (when measuring the length) / perpendicular viewing of the rule or parallax described / start counting from zero / use of a fiducial aid / ruler perpendicular to bench	**1**	**A** repeat **and** average / time a large number of oscillations

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Question	Answer	Marks	Guidance
2(a)(i)	θ = 89(°C)	**1**	correct answer only
2(a)(ii)	s, °C, °C	**1**	**allow** °C seen once **not** C or C° **A** seconds, but not sec(s)
2(b)(i)	12(.0) (°C) and 8(.0) (°C) – both answers correct	**1**	**ecf** (a)(i)
2(b)(ii)	3.0 (°C/min) and 2.0 (°C/min) - both answers correct both answers to 2 significant figures	**1**	**any** significant figures, **ecf** (b)(i) **any** values to 2 significant figures
2(b)(iii)	(expect Q) because the rate of cooling is less / it has a smaller temperature decrease <u>in the same time</u>	**1**	**ecf** (b)(ii)
2(c)	any **two** from: volume of water initial temperature room temperature time (of cooling) size / thickness / material of beaker	**2**	**A** mass / amount
2(d)	remove the lid (from Q) / add a lid to P (and repeat the experiment)	**1**	**A** remove the lid from the other beaker
2(e)	wear goggles / wear gloves (when handling the beaker) / use tongs	**1**	**A** anything sensible **I** do not touch the beaker

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Question	Answer	Marks	Guidance
3(a)(i)	resistor, correct symbol, connected between P and Q	**1**	no line through the symbol
3(a)(ii)	correct parallel connection AND correct symbol	**1**	no line through the symbol
3(b)(i)	V = 3.9 (V)	**1**	correct answer only **not accept** 3.90
3(b)(ii)	I = 0.33 (A)	**1**	correct answer only
3(c)	axes labelled, quantity and unit	**1**	reversed axes loses this mark only
	scales linear, not awkward	**1**	x-axis scale **must** start from 0 scale must fill at least half the grid, no ×3, ×7 etc.
	points plotted accurately, to the nearest ½ square	**1**	**reject** large 'blobs' > ½ square diameter
	thin best-fit straight line drawn	**1**	based on all the plots annotate the graph with ticks / crosses in mark scheme order
3(d)	some indication on the graph of the chosen values	**1**	e.g. triangle drawn / coordinates of chosen points indicated
	⩾ ½ the line between the extreme plotted points used (i.e. $\Delta V \geqslant 0.2$)	**1**	**I** missing negative sign
	G = (–)1.8 – (–)2.2(Ω) inclusive	**1**	**A** 1 significant figure
3(e)	$r = G$ and recorded to the nearest ohm	**1**	
3(f)	connect the voltmeter across / in parallel with the battery terminals (when on open circuit)	**1**	
3(g)	(very) high current flows / cells damaged / wires get (very) hot / insulation melts / battery heats up / circuit overheats / ammeter damaged	**1**	**I** causes a short circuit / damages circuit components / causes electrocution / causes sparks

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Question	Answer	Marks	Guidance
4	**PUT 'SEEN' ON PAGE 14, BECAUSE CANDIDATES MAY WRITE PART OF THEIR ANSWER HERE**		
	MP1: additional apparatus: stopwatch (and measuring cylinder)	**1**	can be awarded from a diagram
	method: **MP2**: (set the fan speed and) measure the initial temperature, final temperature **and** the time taken for the hot water to cool	**1**	
	MP3: repeat for different fan speeds / settings	**1**	
	MP4: control variables: **one** of: initial temperature of the hot water mass / volume / amount of hot water beaker material / size room temperature distance of the beaker from the fan	**1**	all control variables must be stated explicitly and not implied from the method used
	MP5: table: columns to include fan speed / setting, initial temperature, final temperature and time **with correct units**	**1**	**A** plausible symbols instead of words e.g. t, T, θ, θ_I, θ_F **I** any units given for fan speed **A** C and C° and sec(s) here, as these incorrect units have already been penalised in **2(a)(ii)** and also mins/m if previously used and penalised **I** extra columns

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Question	Answer	Marks	Guidance
4	**MP6: processing and conclusion:** **one of**: plot a graph / bar chart of fan speed against rate of cooling plot a graph / bar chart of fan speed against temperature (decrease) plot a graph of temperature against time (for each fan speed) OR compare the readings in the table to see if / how / whether the fan speed affects the (rate of) cooling	1	**I** a prediction e.g., the faster the air speed, the faster the rate of cooling etc. **A** a set of labelled axes drawn – no units needed

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